

NVIDIA Cosmos, Explained in Depth

A beginner-to-intermediate guide to all four generations — Cosmos 1, 2, 2.5, and 3 — with diagrams, model sizes, data, inference speed, and the pretraining + post-training (SFT / GRPO) recipes.

What is a “World Foundation Model” (WFM)?

A **language** model learns from text and predicts the next *word*. A **World Foundation Model** learns from video and predicts what happens next in the *physical world* — how things move, collide, and behave. NVIDIA **Cosmos** is a family of WFMs for **Physical AI** (robots, self-driving cars): instead of testing a robot millions of times in reality, engineers let Cosmos **imagine** realistic video of what would happen and train on that.

Mental model: Cosmos is a “dream engine” for machines. Each generation dreams with higher quality, runs faster, understands your request better, and — by Cosmos 3 — even hears sound and decides physical actions.

Mini-glossary (so the rest makes sense)

Tokenizer	Compresses video into a small set of numbers (“tokens”) the model can handle, and decompresses them back to pixels.
Diffusion	Starts from random noise and repeatedly “cleans” it into a video. High quality, but many steps = slower.
Autoregressive (GPT-style)	Generates the video one chunk at a time, like typing word by word. Fast, good for real-time.
Flow-based	A cleaner, more direct cousin of diffusion: learns a straight “flow” from noise to data, often with fewer steps.
DiT	“Diffusion Transformer” — the Transformer architecture that powers the diffusion engine.
Mixture-of-Transformers	Several specialized Transformers in one model (e.g. one that <i>reasons</i> + one that <i>generates</i>).
Pretraining	Learning general knowledge from a huge, broad dataset.
SFT	Supervised Fine-Tuning: teach the model on curated question→answer examples after pretraining.
GRPO	An RL method: the model tries an answer several times; better-than-average tries get reinforced. Sharpens reasoning.

The big picture: all four generations side by side

	Cosmos 1	Cosmos 2	Cosmos 2.5	Cosmos 3
Released	Jan 2025	Jun 2025	Sep 2025	Jun 2026
Core engine	TWO engines: diffusion + GPT-style	Refined diffusion	ONE unified flow-based model	Mixture-of-Transformers (reason + generate)
Inputs	Text, or image/video	Text, or image/video	Text/image/video (one model)	Text, image, video, AUDIO + ACTIONS
Outputs	Future video	Video (480/704p)	Video up to 30s, multi-camera	Video + actions + audio, with reasoning
Reads text via	T5-XXL encoder	T5-style encoder	Cosmos-Reason1 (VLM)	Built-in reasoning transformer
Model sizes	Diff 7B/14B, GPT 4B/12B	0.6B / 2B / 14B	2B / 14B (+auto/robot variants)	Super / Nano / Edge
Training data	~100M clips (from 20M hrs)	+ action datasets	200M curated clips	~20 trillion tokens
Post-training	Video2World fine-tune	Action-conditioned fine-tune	Model merge + RL	SFT + RL (GRPO-style)
One-word vibe	Kitchen sink	Polish	Unify	Leap

Table 1. Each column is colour-coded and reused throughout this guide: [Cosmos 1](#), [Cosmos 2](#), [Cosmos 2.5](#), [Cosmos 3](#).

How the architecture changed (diagrams)

Cosmos 1 — two parallel engines

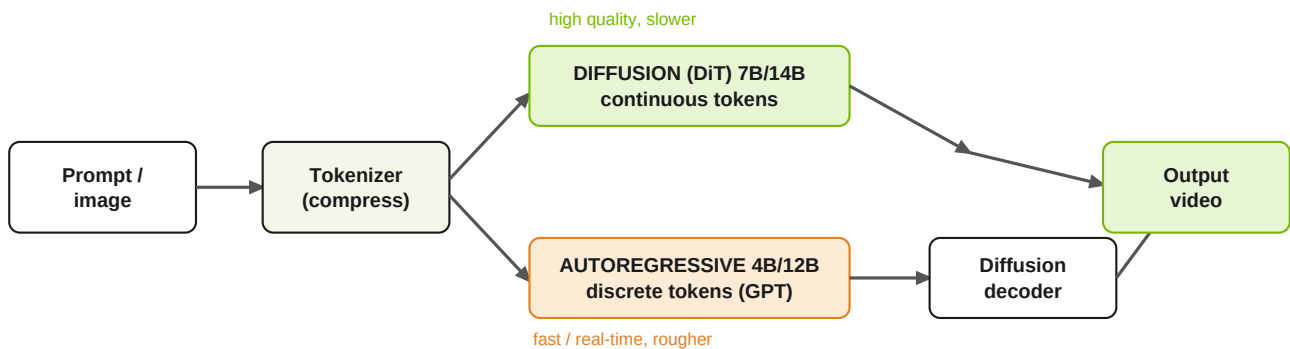


Figure 1. Cosmos 1 shipped two separate engines. Diffusion (top) = best quality; autoregressive (bottom) = fast, but needs a diffusion decoder to clean up its output. Text enters both via a T5-XXL encoder (not shown).

Cosmos 2.5 — one unified flow model with a “smart” text encoder

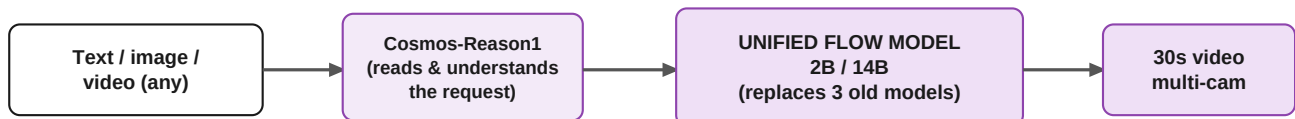


Figure 2. Cosmos 2.5 merged the three separate task-models of earlier versions into a single flow-based model, and upgraded the text encoder to Cosmos-Reason1 — a model that actually understands physics and language.

Cosmos 3 — mixture-of-transformers (thinks before it dreams)

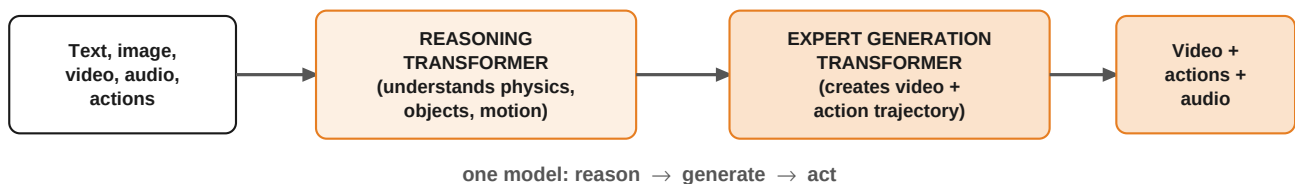


Figure 3. Cosmos 3 is no longer just a video predictor. A reasoning transformer first understands the scene, then an expert generation transformer produces video AND physical actions (and audio) — all in one “omnimodel”.

Deep-dive: why did Cosmos 1 have TWO engines?

This is the most confusing part of Cosmos 1, so it deserves a clear answer. The two engines were **not** a duplicate or a mistake — the paper states it built them on purpose: “We explore two scalable approaches for building pre-trained world foundation models — the diffusion model and the autoregressive model.”

Both solve the **same** hard problem (predict believable future video) with the same trick — break one hard generation into many easy sub-steps — but in two different “spaces”, which forces two different training recipes.

*Analogy: same destination, two routes. Diffusion is a **sculptor** — start from a rough noisy block and refine the whole video over many passes. Autoregressive is a **writer** — produce the video one “token” at a time, left to right, like an LLM.*

Aspect	Diffusion engine (sculptor)	Autoregressive engine (writer)
Token space	Continuous latents (CV 8×8×8 tokenizer)	Discrete codes (DV 8×16×16; 64,000-word visual vocabulary)
Learning signal	Denoising score-matching (EDM): recover the clean signal from a noised one	Next-token cross-entropy: predict the next code (exactly like an LLM)
Text conditioning	T5-XXL via cross-attention	T5 via cross-attention added to blocks
Training stages	Text2World pretrain → Video2World fine-tune (add observed frames)	Video-only next-token pretrain → add text for Video2World
Model sizes	7B / 14B	4B / 12B (5B / 13B for Video2World)
Strength	Highest visual fidelity	Speed + streaming; plugs into the LLM toolbox (KV-cache, real-time)
Weakness	Many denoising steps → slower	Heavy compression → artifacts → needs a diffusion decoder to repaint detail

Table 2a. The two engines use different tokenizers, different losses, and different training stages — they are genuinely different models, not one model trained twice.

So why keep both?

They are **complementary bets**. Diffusion is the **quality** champion; autoregressive is the **speed + LLM-integration** champion (it treats video like language — exactly the direction Cosmos 3 later embraced). The **diffusion decoder** exists only because the autoregressive path's aggressive discrete compression “can sometimes lead to undesired distortions” — so a small diffusion model cleans its output back up.

Why each new generation? (the problem it solved)

A fair question: was each new version just a **bigger model on more data**? **No**. Scale grew every time, but it was the *enabler*, not the headline — each step introduced a distinct architectural or algorithmic idea to fix a concrete problem.

Step	Key problem with the previous version	How they solved it	Real novelty (beyond size/data)
1 → 2	Too slow, too many separate engines, visible hallucinations — a research platform, not a daily tool.	Commit to the diffusion path; add sparse attention; engineer better quality/control; add small fast variants; add action-conditioned post-training.	Sparse attention ($\approx 2.6\times$ faster) + action conditioning.
2 → 2.5	Three separate task-models to juggle; a T5 text encoder that doesn't "understand" physics; short, single-camera video.	Unify the 3 tasks into ONE flow-based model; replace T5 with the Cosmos-Reason1 VLM as encoder; add RL post-training + model merging; extend to 30s, multi-camera.	Task unification + flow matching + reasoning-VLM encoder + RL.
2.5 → 3	Still only a video predictor — can't natively reason, hear, or act; understanding and generation were separate models.	One mixture-of-transformers omnimodel: a reasoning transformer + an expert generation transformer; reasons before it generates; adds audio + action.	New architecture (MoT) + new modalities (audio/action) + merging understand/generate/act.

Table 2b. Read the rightmost column: every jump is a genuine idea, not a scale-up. 1→2 buys speed; 2→2.5 buys understanding + simplicity; 2.5→3 buys a whole new capability.

The one-sentence “why”

- **1→2:** “Make it fast and reliable enough to actually use.”
- **2→2.5:** “Make it one model that truly understands the request.”
- **2.5→3:** “Stop just watching the world — reason about it, hear it, and act in it.”

The data: scale and diversity

A WFM is only as good as the video it learns from. Cosmos 1 curated **~100 million clips** out of **20 million hours** of raw video, deliberately balanced across nine kinds of physical scene so the model isn't biased toward one (e.g. only driving). Later generations grew the data by orders of magnitude.

Cosmos 1 training-data mix (by category)

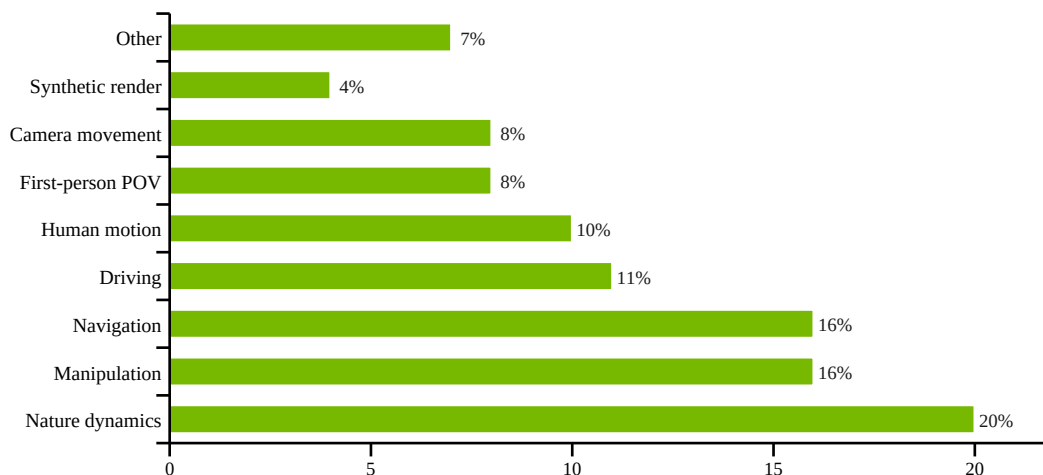


Figure 4. Diversity is intentional: nature, robot manipulation, and navigation dominate, but driving, human motion and synthetic data are all represented. This breadth is what lets one pretrained model be fine-tuned for many downstream robots/vehicles.

Data scale across generations

Generation	Pretraining data	Notes
Cosmos 1	~100M clips (from ~20M hours)	9 balanced categories; ~30% removed by dedup
Cosmos 2	Cosmos 1 data + action datasets	Bridge, AgiBotWorld, GR00T Dreams for post-training
Cosmos 2.5	200M curated high-quality clips	+ domain post-training data (auto, robotics)
Cosmos 3	~20 trillion tokens	~1B images, ~400M videos, + audio + action trajectories

Table 2. The jump from “100M clips” to “20T tokens” reflects Cosmos 3 also ingesting text, audio and robot/human action data — not just video. **Caveat:** Cosmos 3’s exact counts (20T tokens, ~1B images, ~400M videos) come from press coverage; NVIDIA’s official release states only “billions of samples across text, image, video, sound and action.”

The key insight: each generation adds a new TYPE of data

The data story is **qualitative**, not just “more”. Each generation introduces a new *kind* of data, and that new kind is exactly what unlocks the next capability:

- **Cosmos 1 — broad video** (9 balanced categories) → learns **general physics**.
- **Cosmos 2 — + action-labelled robot data** (Bridge, AgiBotWorld, GR00T Dreams) → learns **“this action causes that outcome.”**
- **Cosmos 2.5 — + multi-camera / domain data** (7-cam driving, 3-cam robotics) → learns **deployment realism and multi-view consistency**.
- **Cosmos 3 — + audio and action trajectories** (from humans and robots) → enables the **omnimodel** that can hear and act, not just see.

Model sizes & key hyperparameters

“Parameters” (B = billion) are the model's adjustable knobs — more usually means smarter but slower/heavier. Cosmos offers small *and* large versions so you can trade quality for speed.

Every model variant at a glance

Generation	Variants (parameters)	Helper models
Cosmos 1	Diffusion 7B & 14B; Autoregressive 4B & 12B (5B/13B for Video2World)	Tokenizer 77–105M; Prompt-up sampler 12B; Diffusion-decoder 7B
Cosmos 2	0.6B (Text2Image), 2B, 14B	Sparse-attention inference path
Cosmos 2.5	2B & 14B (each: pre-trained + post-trained); auto 7-cam & robot 3-cam variants	Cosmos-Reason1 as text encoder
Cosmos 3	Super (max accuracy), Nano (sub-second), Edge (local, soon)	Built-in reasoning transformer

Cosmos 1 diffusion hyperparameters (the most fully documented)

Setting	7B model	14B model
Transformer layers	28	36
Model (hidden) dimension	4,096	5,120
Attention heads	32	40
AdaLN-LoRA rank	256	256
Learning rate	2 ⁻¹⁵	2 ⁻¹⁶
Optimizer	AdamW (beta1=0.9, beta2=0.99), weight decay 0.1–0.2	
Warmup	2,500 iterations, linear	
Text encoder	T5-XXL, 512 tokens via cross-attention	
Resolution schedule	512p (57 frames) → 720p (121 frames)	
Context length @720p	56,320 tokens	
Diffusion formulation	EDM denoising score-matching; QK-RMSNorm for stability	
Compute	~10,000 H100 GPUs, ~3 months	

Table 3. NVIDIA published far more hyperparameters for Cosmos 1 than for the closed-er later releases; 2.5 and 3 reuse the same design DNA (3D-RoPE positions, AdaLN-LoRA, a causal tokenizer) at larger scale.

Inputs and outputs: the “data contract” (DTO)

Think of a **DTO** (data-transfer object) as the exact shape of what you hand the model and what it hands back. This shape widened a lot across generations.

Generation	INPUT (what you provide)	OUTPUT (what you get back)
Cosmos 1 (diffusion)	Text → T5-XXL embedding (512 tokens) + optional conditioning frames + fps + frame-count	Continuous latent video → decoded to RGB frames
Cosmos 1 (autoregressive)	Discrete video tokens (DV 8×16×16) + T5 text embedding	Next discrete tokens → diffusion-decoder → RGB frames
Cosmos 2	Text, or image, or video + fps	RGB video at 480p / 704p, 10–16 fps
Cosmos 2.5	Text / image / video, encoded by Cosmos-Reason1	Up to 30s video; optional multi-camera (3–7 synced views)
Cosmos 3	Text + image + video + ambient audio + action trajectories	Video + action trajectory + audio, preceded by explicit reasoning

Table 4. The input distribution broadens from “text + frames” (Cosmos 1) to a full five-modality stream including audio and actions (Cosmos 3). The output likewise grows from “a short video” to “video + the physical actions a robot should take”.

Why the input distribution matters

- **Cosmos 1–2:** trained mostly on curated *internet-style* video across 9 categories — great general physics, but you fine-tune for your specific robot.
- **Cosmos 2.5:** 200M *physical-AI-focused* clips + domain post-training (driving, manipulation) shift the distribution toward real deployment.
- **Cosmos 3:** adds *action trajectories from real humans and robots* and *audio* to the training mix — so the input/output distribution now includes the robot's own behaviour, not just what a camera sees.

Key results: inference speed

Speed is the headline practical difference. Faster generation = more simulations per day = faster robot/AV development. Each generation attacked latency differently.

Generation	Speed lever	Reported result
Cosmos 1	Causal, factorized tokenizer	Up to 12× faster encode/decode vs. prior tokenizer (CogVideoX)
Cosmos 2	Sparse attention in the DiT	Up to 2.6× end-to-end inference speedup
Cosmos 2.5	Flow-based sampling (fewer steps) + smaller 2B option	Faster, more resource-efficient than 2.x diffusion
Cosmos 3	Dedicated Nano variant	High-quality video + action reasoning in a fraction of a second

Table 5. Two concrete, comparable speedups are published: Cosmos 1's tokenizer (up to 12×) and Cosmos 2's sparse attention (up to 2.6×). 2.5 and 3 emphasise efficiency via flow sampling and a sub-second Nano tier rather than a single headline multiplier.

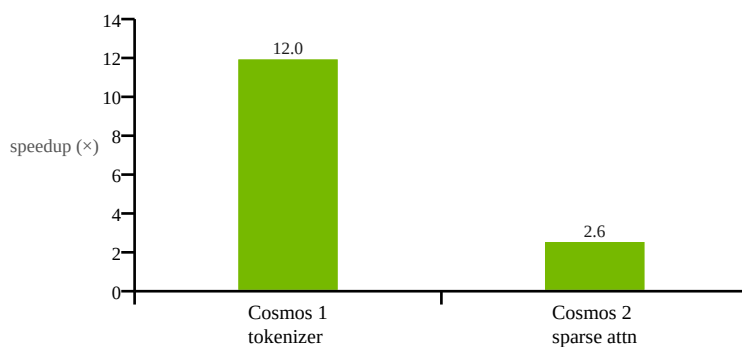


Figure 5. The two directly-quoted “× speedup” numbers. (Different things are being sped up — tokenizer vs. whole pipeline — so this compares *magnitude of improvement*, not absolute latency.)

The quality/speed trade-off in one sentence

Diffusion (Cosmos 1–2) = many denoising steps = top quality but slower; **autoregressive** (Cosmos 1) = stream tokens = fast but rougher; **flow-based** (Cosmos 2.5) = fewer steps for similar quality; **Nano** (Cosmos 3) = engineered for sub-second responses.

Zoom-in: what is “sparse attention”? (the Cosmos 2 trick)

Cosmos 2's headline speedup (up to 2.6×) comes from **sparse attention**. Here is the idea with no maths.

The problem: “attention” is very expensive for video

Inside a Transformer, every token (think: every little patch of the video) **looks at every other token** to decide what is relevant. That is called **full attention**. A short 720p video is **tens of thousands of patches** (frames × height × width), and full attention compares *every patch with every other patch* — so the cost grows **quadratically**: double the patches → **4×** the work. For video, this one step dominates the whole generation time.

The insight: most of those comparisons are wasted

A patch in the top-left of frame 1 has almost nothing to do with a patch in the bottom-right three seconds later. What actually matters to a patch is its **neighbours** — nearby in space, and a frame or two before/after.

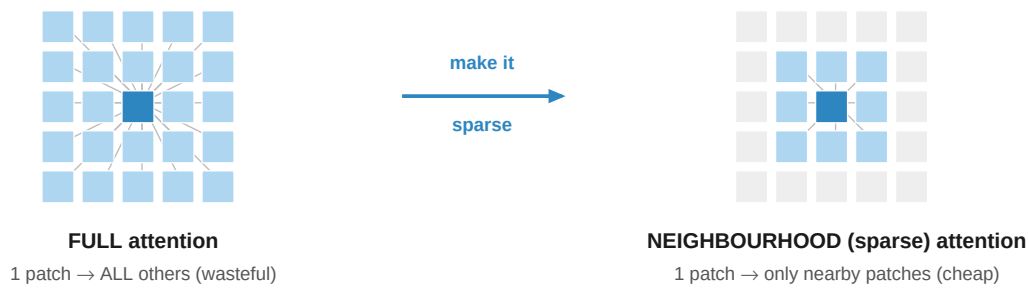


Figure 5a. Left: every patch attends to all others (lines everywhere). Right: each patch attends only to its local window — the grey patches are skipped. Cosmos 2 (via the NATTEN library) drops up to 98% of the connections, keeping only the ~2% that matter.

The fix: Neighbourhood Attention (NATTEN)

Instead of attending to **all** patches, each patch attends only to those in a small **local window** around it (in space and time). Cosmos 2 pushed this so **up to 98% of the attention connections are dropped** (sparsity raised “from 50% to 98%”) — only the most relevant ones are computed.

- **Result:** far fewer calculations → **1.7×–2.6× faster** at 720p, with quality essentially preserved (the dropped links were the unimportant ones).
- **Bonus:** it is an inference-time efficiency change, not a bigger model — related “Generalized Neighbourhood Attention” can even be plugged into existing models for 28–46% speedups **without any retraining**.

Analogy: full attention = everyone in a stadium trying to talk to everyone else at once (chaos, won't scale). Neighbourhood attention = each person only talks to their own row and the rows just ahead/behind — you keep all the context that matters, and it scales.

Pretraining and post-training: SFT and GRPO

Modern Cosmos models are built in two phases. **Pretraining** soaks up broad knowledge from massive data. **Post-training** then sharpens behaviour with curated examples (**SFT**) and reinforcement learning (**GRPO**). This recipe is clearest in **Cosmos-Reason1** — the reasoning brain that powers Cosmos 2.5's text understanding and the spirit of Cosmos 3.

The 4-stage training pipeline (Cosmos-Reason1)

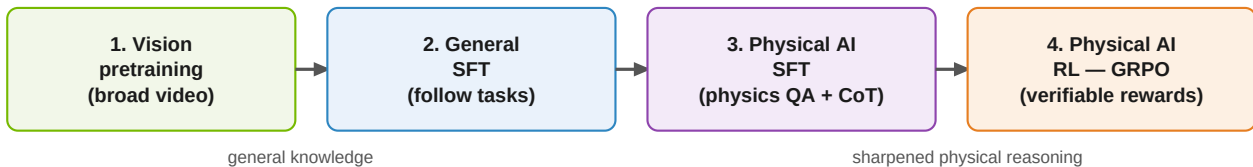


Figure 6. Knowledge first (stages 1–2), then physical-world specialisation (stage 3), then reinforcement learning to make the reasoning reliable (stage 4).

What happens in SFT (stage 3)

- Curated **question** → **answer** examples for physical common sense (space, time, basic physics) and embodied decisions (what should the robot do?).
- Built from real robotics/AV datasets: **BridgeData V2**, **RoboVQA**, **AgiBot**, **HoloAssist**, and driving data.
- The team hand-picked **~200–250 examples per data source**, turned them into multiple-choice questions, and removed ambiguous ones — quality over quantity.
- Answers include **chain-of-thought** (the model explains its reasoning step by step).

What happens in GRPO (stage 4) — in plain words

GRPO (Group Relative Policy Optimization) is reinforcement learning *without* a separate “judge” model. For each question the model produces **several attempts**; because these questions have a **verifiable correct answer**, each attempt can be auto-graded. Attempts that score **above the group average** are reinforced; below-average ones are discouraged. A **reference model** and **reward normalization** keep the updates stable so the model improves without “drifting”. The payoff: noticeably better physical reasoning.

Key experimental results

Cosmos-Reason1: bigger model & RL both help

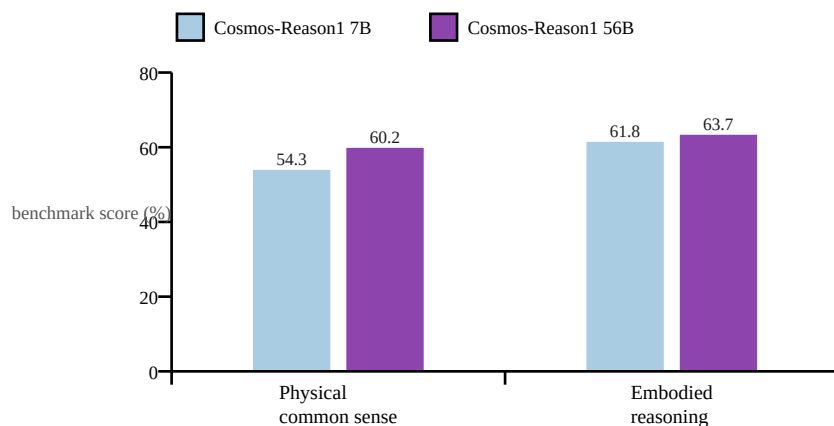


Figure 7. The 56B model beats the 7B on both physical-common-sense and embodied-reasoning benchmarks. Embodied reasoning improved by ~+10–11 points over the next-best prior model. **Source note:** numbers are from NVIDIA's Cosmos-Reason1 research page (released 7B checkpoint). An earlier arXiv v1 of the paper reported an 8B model with somewhat different scores — see the verification page.

Reinforcement learning (GRPO) lifts reasoning further

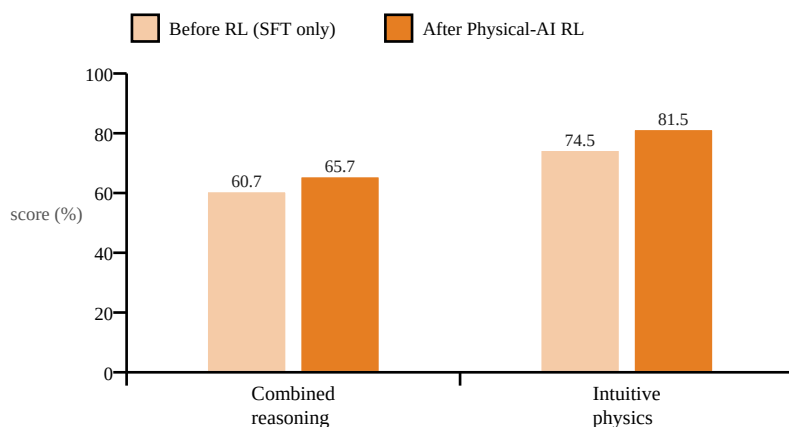


Figure 8. Adding the GRPO reinforcement-learning stage raised the 7B model by +5.0 points (combined reasoning) and +7.0 points (intuitive physics) — evidence that post-training, not just size, drives physical reasoning. **Source note:** research-page figures; arXiv v1 reports different magnitudes (e.g. intuitive physics 65.7→68.7). The *direction* — RL helps — is consistent across both.

Generation-quality results (Cosmos 2.5 & 3)

- **Cosmos 2.5** post-trained **14B** scores **0.810** on PAI-Bench Image2World (matching the strong Wan2.2-A14B baseline).
- **Cosmos 2.5** autonomous-driving multiview model: up to **2.3×** better FVD/FID (video realism metrics) versus its predecessor.
- **Cosmos 3** ranks **#1 among open models** on Artificial Analysis, Physics-IQ, PAI-Bench and R-Bench (world generation); RoboLab and RoboArena (action policy); and VANTAGE-Bench and TAR (vision understanding).

Note: FVD/FID measure how realistic/consistent generated video is (lower is better, so a 2.3× improvement is large). PAI-Bench scores world-generation quality (higher is better).

Verification & source audit

Every quantitative claim in this guide was cross-checked against primary sources (NVIDIA papers, research pages, and the official Cosmos 3 release). Of 13 checked claims, **11 verified exactly**; **2 carry source-version or disclosure caveats** (marked “!”). No factual errors were found in any claim that could be checked against a primary source.

#	Claim in this guide	Checked against	Verdict
1	Cosmos 1 data mix: 20/16/16/11/10/8/8/4/7%	arXiv 2501.03575	OK — exact
2	~100M clips (10^8) from ~20M hours of video	arXiv 2501.03575	OK — exact
3	Diffusion 7B = 28L / 4096 / 32 heads; 14B = 36L / 5120 / 40	Table 11	OK — exact
4	AdaLN-LoRA rank 256; LR 2^-15 / 2^-16; context 56,320	arXiv 2501.03575	OK — exact
5	Autoregressive 4B / 12B (+ 5B / 13B Video2World)	arXiv 2501.03575	OK — exact
6	Trained on ~10,000 H100 GPUs for ~3 months	arXiv 2501.03575	OK — exact
7	Tokenizer up to 12× faster than prior (CogVideoX)	arXiv 2501.03575	OK
8	Cosmos 2: 0.6B/2B/14B; sparse attn up to 2.6×; 480/704p	GitHub + NVIDIA blog	OK
9	Cosmos 2.5: flow; Reason1 encoder; 2B/14B; 200M clips; 30s; PAI-Bench 0.810; 2.3× FVD/FID	NVIDIA research page	OK
10	GRPO post-training; 4-stage pipeline; verifiable rewards	arXiv 2503.15558	OK
11	Cosmos-Reason1 = 7B & 56B with the Fig 7–8 scores	Research page vs arXiv v1	CAVEAT — see below
12	Cosmos 3: mixture-of-transformers omnimodel; Super/Nano/Edge; #1 on open leaderboards	NVIDIA newsroom (Jun 2026)	OK
13	Cosmos 3 data = 20T tokens, ~1B images, ~400M videos	Press coverage (not NVIDIA)	CAVEAT — see below

Table 6. Claims audit. Green = verified against a primary source; amber = caveat.

The two caveats, in detail

- **! Cosmos-Reason1 size & scores (claim 11):** NVIDIA's research page describes a released **7B** model with the scores plotted in Figures 7–8. The **arXiv v1** of the paper instead describes an **8B** model and reports different numbers (e.g. physical common sense 52.3% vs 54.3%; intuitive-physics RL gain 65.7→68.7 vs 74.5→81.5). This is a paper-revision difference. This guide uses the **research-page (7B)** figures because that matches the publicly released checkpoint — but treat the exact decimals as version-dependent. The qualitative conclusions (bigger model helps; RL helps) hold in both.
- **! Cosmos 3 data scale (claim 13):** the “20 trillion tokens / ~1B images / ~400M videos” figures come from **press coverage** of the launch. NVIDIA's own newsroom release only says “billions of samples across text, image, video, sound and action.” Treat the precise counts as **unconfirmed by NVIDIA**.

Bottom line: the guide is accurate where primary sources exist. The only soft spots are (a) decimal-level benchmark numbers for Cosmos-Reason1, which differ between paper versions, and (b) Cosmos 3's exact data counts, which NVIDIA has not officially published.

Putting it all together

The four generations trace a clear arc:

- **Cosmos 1 (Toolbox):** two engines (diffusion + GPT), a strong causal tokenizer, T5 text, 9-category data, fully documented hyperparameters. Maximum flexibility.
- **Cosmos 2 (Polish):** diffusion-only, sparse attention for up to 2.6× speed, action-conditioned post-training. Same idea, executed better.
- **Cosmos 2.5 (Unify):** one flow-based model replaces three; Cosmos-Reason1 reads the prompt; 200M clips; 30s multi-camera video; RL + model-merge post-training.
- **Cosmos 3 (Leap):** mixture-of-transformers omnimodel; reasons before generating; adds audio and actions; ~20T-token training; Super/Nano/Edge tiers; #1 open model on many leaderboards.

The single sentence to remember

Cosmos went from **“many tools that dream video”** (1) → **“the best tool, polished”** (2) → **“one smart unified tool”** (2.5) → **“a reasoning, seeing, hearing, acting omnimodel”** (3).

Sources: NVIDIA *Cosmos World Foundation Model Platform* (arXiv:2501.03575); *Cosmos-Reason1: From Physical Common Sense to Embodied Reasoning* (arXiv:2503.15558); NVIDIA Research pages for Cosmos-Predict2.5 and Cosmos-Reason1; NVIDIA developer blogs for Cosmos-Predict2; Hugging Face NVIDIA blog on Cosmos Predict/Transfer 2.5; NVIDIA newsroom release for Cosmos 3 (Jun 2026). Figures are illustrative; numbers quoted from these sources. Educational summary.